📅 DAY 2 – Networking Basics (IP, DNS,Ports)

🎯 Objective

- Understand what a network is
- Learn IP address basics
- Understand DNS and how websites work
- Learn ports and protocols
- Run basic network commands

🌐 PART 1: What is a Network?

A network is a group of two or more computers or devices connected together to share data, resources, and communication.

Examples of a Network:

- Internet
    - Wi-Fi at home
    - College computer lab
    - Office computers connected together

🧠 PART 2: IP Address

An **IP Address (Internet Protocol Address)** is a **unique address given to a device** on a network. It helps devices **identify and communicate** with each other.

Example of an IP Address:

- 192.168.1.1
- 8.8.8.8
- 142.250.x.x

Why is an IP Address Important?

An IP address is used to:

- Identify a device on a network
- Send data to the correct device
- Access websites and internet services
- Connect computers, phones, routers, and servers

Types of IP Address:

1) Private IP Address

Used inside a local network like:

- Home Wi-Fi
- College lab
- Office network

Example:

- 192.168.1.10

2) Public IP Address

Used on the internet and visible outside your local network.

Example:

- 142.250.x.x

Ports and Protocols

What is a Port?

A **port** is a **communication endpoint** in a computer or network.

Simple Meaning:

👉 A port is like a **door number** in a house.

- **IP Address** tells **which house**
- **Port number** tells **which door**

Example

When you open a website, the data goes to a specific port.

For example:

- Web browsing uses **Port 80** or **Port 443**
- Remote login uses **Port 22**

What is a Protocol?

A **protocol** is a **set of rules** that devices follow to communicate over a network.

Simple Meaning:

👉 A protocol is like a **language or rule book** for communication.

It tells devices:

- How to send data
- How to receive data
- How to understand the message

Basic Network Commands

Basic network commands are used to **check connectivity, find IP addresses, and troubleshoot network problems**.

1) ping Command

Purpose

The ping command is used to **check whether a device or website is reachable** over the network.

Syntax

ping google.com

2) nslookup Command

Purpose

The nslookup command is used to **find the IP address of a website**.

Syntax

nslookup google.com

3) traceroute / tracert Command

Purpose

This command is used to **trace the path taken by data packets** from your computer to a destination website or server.

Syntax

traceroute google.com